Abstract vector spaces

The final step: vectors don't have to be arrows.

THE IDEA

Everything in this series worked with arrows and lists of numbers — but nothing in the reasoning required them. If a set of objects can be **added** and **scaled** sensibly, all of linear algebra applies. Such a set is a **vector space**, and its members are vectors regardless of what they look like.

These are all vectors	Because you can…
Arrows in the plane	add them tip-to-tail, scale their length
Lists of numbers	add componentwise, multiply by a scalar
Functions	add two functions, multiply a function by a number
Polynomials	same — and they have a basis: $1, x, x^2, x^3$ …

> **The punchline:** the **derivative is a linear transformation**. Written in the basis $1, x, x^2, x^3$, differentiation becomes an ordinary matrix — so calculus can be done with matrix multiplication.

WORKED EXAMPLE

CHAPTER 16 — WORKED EXAMPLE

vectors do not have to be arrows

THE IDEA

anything you can ADD and SCALE sensibly is a vector :

arrows · lists of numbers · functions · polynomials

EXAMPLE — the derivative is a MATRIX

basis of polynomials : $1,\ x,\ x^2,\ x^3$

$p(x) = 3 + 2x + 5x^2$ → coords $(3, 2, 5, 0)$

$$\text{derivative matrix } D = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

apply D to (3, 2, 5, 0) :

row 1 : $0(3) + 1(2) + 0(5) + 0(0) = 2$

row 2 : $0(3) + 0(2) + 2(5) + 0(0) = 10$

rows 3, 4 $= 0$

result (2, 10, 0, 0) → 2 + 10x

and indeed $d/dx\,(3 + 2x + 5x^2) = 2 + 10x$ ✓

WHY THE ABSTRACTION PAYS OFF

Because the rules were never about arrows, every theorem you have learned applies anywhere the rules hold — function spaces, probability distributions, quantum states, and the high-dimensional spaces where machine learning lives. That generality is the real reason linear algebra is everywhere.

TEST — Chapter 16

1. In the basis $1, x, x^2, x^3$, write the coordinates of $p(x) = 4 + 0x + 2x^2$. **→ (4, 0, 2, 0)**
2. Apply the derivative matrix to (4, 0, 2, 0) and give the resulting polynomial. **→ (0, 4, 0, 0) → 4x**
3. Is the set of all 2×2 matrices a vector space? Justify in one line. **→ Yes — you can add two matrices and scale a matrix, so the rules hold**

FORMULAS FROM THIS CHAPTER

Vector space	any set you can add and scale
Polynomial basis	$1, x, x^2, x^3 \ldots$
Key fact	the derivative is a linear transformation